# Marcus W. Fedarko

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## Summary


I'm a postdoc at the University of Maryland. I develop computational methods to help people study microbiome sequencing data in high resolution, with foci in metagenome assembly and data visualization.


## Education

**Ph.D., Computer Science** 9/2018–6/2025
University of California, San Diego La Jolla, CA

**M.S., Computer Science** 9/2018–6/2022
University of California, San Diego La Jolla, CA

**B.S. with High Honors, Computer Science** 9/2014–5/2018
University of Maryland College Park, MD

## Research Experience

**Postdoctoral Associate** 10/2025–Present
University of Maryland College Park, MD

**Graduate Student Researcher** 9/2018–10/2025
University of California, San Diego La Jolla, CA

**Research Intern** 6/2016–8/2018
University of Maryland College Park, MD

## Teaching Experience

**Graduate Teaching Assistant**
University of California, San Diego La Jolla, CA

- **CSE 185:** Advanced Bioinformatics Laboratory 3/2025–6/2025
- **CSE 181:** Molecular Sequence Analysis 1/2025–3/2025, 1/2024–3/2024
- **CSE 282:** Introduction to Bioinformatics Algorithms 1/2025–3/2025, 1/2023–3/2023, 1/2022–3/2022, 1/2021–3/2021

**Course Assistant**
Marine Biological Laboratory Woods Hole, MA

- **STAMPS:** Strategies and Techniques for Analyzing Microbial Population Structures 7/2018–8/2018
- **MOLE:** Workshop on Molecular Evolution 7/2018

**Undergraduate Teaching Assistant**
University of Maryland College Park, MD

- **CMSC 330:** Organization of Programming Languages 8/2016–12/2016

## Other Programming / Writing Experience

**Software Developer** 7/2025–10/2025
*Bioinformatics Algorithms* (Compeau & Pevzner) Remote

**Student Staff Writer** 1/2015–9/2017
University of Maryland Department of Computer Science College Park, MD

**Student Intern** 5/2013–8/2014
Axiometric Columbia, MD

**Intern Software Engineer** 7/2012–8/2012
Battlefield Telecommunications Systems Columbia, MD

## Peer-Reviewed Papers

8. Zhang Z, **Fedarko MW**, Bankevich A, and Pevzner PA (2026). "MGA: a tool for haplotype-mixed assembly of long and accurate reads." *Genome Biology, 27*(1):201.

7. **Fedarko MW**, Kolmogorov M, and Pevzner PA (2022). "Analyzing rare mutations in metagenomes assembled using long and accurate reads." *Genome Research, 32*(11-12):2119–2133.

6. Cantrell K*, **Fedarko MW***, Rahman G, McDonald D, Yang Y, Zaw T, Gonzalez A, Janssen S, Estaki M, Haiminen N, Beck KL, Zhu Q, Sayyari E, Morton JT, Armstrong G, Tripathi A, Gauglitz JM, Marotz C, Matteson NL, Martino C, Sanders JG, Carrieri AP, Song SJ, Swafford AD, Dorrestein PC, Andersen KG, Parida L, Kim H-C, Vázquez-Baeza Y, and Knight R (2021). "EMPress Enables Tree-Guided, Interactive, and Exploratory Analyses of Multi-omic Data Sets." *mSystems, 6*(2):e01216-20. (* = contributed equally)

5. Huey SL, Jiang L, **Fedarko MW**, McDonald D, Martino C, Ali F, Russell DG, Udipi SA, Thorat A, Thakker V, Ghugre P, Potdar RD, Chopra H, Rajagopalan K, Haas JD, Finkelstein JL, Knight R, and Mehta S (2020). "Nutrition and the Gut Microbiota in 10- to 18-Month-Old Children Living in Urban Slums of Mumbai, India." *mSphere, 5*(5):e00731-20.

4. **Fedarko MW**, Martino C, Morton JT, González A, Rahman G, Marotz CA, Minich JJ, Allen EA, and Knight R (2020). "Visualizing 'omic feature rankings and log-ratios using Qurro." *NAR Genomics and Bioinformatics, 2*(2):lqaa023.

3. Sanders JG, Nurk S, Salido RA, Minich J, Xu ZZ, Martino C, **Fedarko M**, Arthur TD, Chen F, Boland BS, Humphrey GC, Brennan C, Sanders K, Gaffney J, Jepsen K, Khosroheidari M, Green C, Liyange M, Dang JW, Phelan VV, Quinn RA, Bankevich A, Chang JT, Rana TM, Conrad DJ, Sandborn WJ, Smarr L, Dorrestein PC, Pevzner PA, and Knight R (2019). "Optimizing sequencing protocols for leaderboard metagenomics by combining long and short reads." *Genome Biology, 20*(1):226.

2. Ghurye J, Treangen T, **Fedarko M**, Hervey WJ, and Pop M (2019). "MetaCarvel: linking assembly graph motifs to biological variants." *Genome Biology, 20*(1):174.

1. Meisel JS, Nasko DJ, Brubach B, Cepeda-Espinoza V, Chopyk J, Corrada-Bravo H, **Fedarko M**, Ghurye J, Javkar K, Olson ND, Shah N, Allard SM, Bazinet AL, Bergman NH, Brown A, Caporaso JG, Conlan S, DiRuggiero J, Forry SP, Hasan NA, Kralj J, Luethy PM, Milton DK, Ondov BD, Preheim S, Ratnayake S, Rogers SM, Rosovitz MJ, Sakowski EG, Schliebs NO, Sommer DD, Ternus KL, Uritskiy G, Zhang SX, Pop M, and Treangen TJ (2018). "Current progress and future opportunities in applications of bioinformatics for biodefense and pathogen detection: Report from the Winter Mid-Atlantic Microbiome Meet-up, College Park, MD January 10$^{th}$, 2018." *Microbiome,*

*6*(1):197.

## Open-Source Software

6. **wotplot:** Library for creating and visualizing dot plot matrices.
   https://github.com/fedarko/wotplot
5. **strainFlye (☕$_7$):** Pipeline for the analysis of rare mutations in metagenomes.
   https://github.com/fedarko/strainFlye
4. **EMPress (☕$_6$):** Visualization tool for phylogenetic trees and associated data.
   https://github.com/biocore/empress
3. **Qurro (☕$_4$):** Visualization tool for log-ratios of compositional data.
   https://github.com/biocore/qurro
2. **pyfastg:** Library for parsing FASTG-format assembly graph files.
   https://github.com/fedarko/pyfastg
1. **MetagenomeScope:** Visualization tool for metagenome assembly graphs.
   https://github.com/marbl/MetagenomeScope

Projects marked (☕$_n$) represent the "main contribution" of the $n$-th peer-reviewed paper in the list above.

## Talks

6. "Hierarchical decomposition and visualization of metagenome assembly graphs."
   *ISMB (Intelligent Systems for Molecular Biology) BioVis 2026.* 7/2026
   *CAVE (Chesapeake Aquatic Viral Ecology) 2026.* 4/2026
   *M$^3$ (Mid-Atlantic Microbiome Meet-up) 2026.* 3/2026
5. "Metagenome assembly." 1/2024
   *Guest lecture for ESE 184 (Computational Tools for Decoding Microbial Ecosystems), California Institute of Technology.*
4. "Studying microbiomes using DNA sequencing." 5/2023
   *Presentation to students from Kearny High School visiting UC San Diego.*
3. "EMPress: a tool for tree-guided exploratory analysis of multi-omic datasets." 1/2021
   *UC San Diego CMI Connect Webinar.*
2. "Visualizing, exploring, and understanding microbiome sequencing data." 1/2020
   *UC San Diego CSE Research Open House.*
1. "Visualizing metagenomic assembly graphs, doing undergrad research at UMD, applying to grad schools, and probably other stuff along the way." 4/2018
   *Guest lecture for CMSC 396H (Undergraduate Honors Seminar), University of Maryland.*

## Editing and Reviewing

2. **Textbook Editor:** *Advanced Bioinformatics Algorithms* 2024
1. **Peer Reviewer:** *Bioinformatics Advances*, *PLOS ONE* Ad hoc

## Service

8. **Volunteer:** Maryland Day "Bites and Bytes" booth 4/2026

7. **Organizer:** "Algorithmic Bioinformatics" seminar series 12/2025–
6. **Panelist:** CSE Society at UCSD × Triton Software Engineering research panel 5/2025
5. **System Administrator:** Pevzner Lab computing server 2/2023–10/2025
4. **Mentor:** UC San Diego Graduate Women in Computing mentorship program 10/2021–6/2024
3. **Moderator:** QIIME 2 forum (https://forum.qiime2.org) 3/2020–4/2024
2. **Co-organizer:** UC San Diego CSE Visit Day 1/2019–3/2024
1. **Code Review (Co-)organizer:** Knight Lab 12/2018–8/2020

## Honors and Awards

10. **Chez Bob Hackathon Award:** UC San Diego 2024
9. **CMNS Dean's List:** University of Maryland 2014–2018
8. **University Honors Citation:** University of Maryland 2017
7. **Rita Colwell Travel Fellowship:** University of Maryland 2017
6. **Travel Award, EGS Workshop:** University of Michigan 2017
5. **John D. Gannon Endowed Scholarship:** University of Maryland 2017
4. **Corporate Partners in Computing Scholarship:** University of Maryland 2016, 2017
3. **Member:** Omicron Delta Kappa National Leadership Honor Society 2016
2. **Scholarship for Employees' Children:** Northrop Grumman 2014
1. **Dean's Scholarship:** University of Maryland 2014

## Skills

- **Languages:** English (native), German (basic), Hindi (basic)
- **Computer Languages:** Python, JavaScript, HTML / CSS, LaTeX, Java, C, C++, ...
- **Software Libraries:** matplotlib, pandas, NumPy, SciPy, scikit-bio, Dash, Cytoscape.js, ...